MULTI MODAL AI TUTORING SYSTEM

Project ID: 25-26J-130

Project Proposal Report

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August 2025

Title Page

- Title of the Project: Subcomponent – Agentic AI Core
- Project ID: 25-26J-130
- Document: Project Proposal Report
- Name of the Authors: Dilshan B A T
- Official Name of the Degree: BSc (Hons) in Information Technology Specialized in Data Science
- Official Name of the Department, Faculty, and the University: Department of Computer Science, Faculty of Computing, Sri Lanka Institute of Information Technology
- Month and Year of Submission: August 2025

Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

This proposal presents the Agentic AI Core, a central reasoning module for a multimodal intelligent tutoring system. Existing tutoring systems largely operate reactively, responding only after student errors occur [1]. In contrast, the Agentic AI Core introduces proactive, goal-oriented planning by simulating the behavior of a human tutor. It uses memory-based goal tracking and advanced planning frameworks such as Tree of Thought and ReAct to break long-term learning objectives into adaptive instructional sequences [2], [3].

By continuously monitoring learner progress, emotional state, and engagement - gathered from integrated Computer Vision and Reinforcement Learning modules - the Agentic AI Core dynamically updates its strategy. This allows it to personalize the teaching path in real time, enhancing coherence, foresight, and responsiveness. Unlike current systems which rely on fixed rules or short-term decisions, this component provides strategic guidance, real-time adaptation, and pedagogical memory, enabling more human-like tutoring behaviors. Its novelty lies in integrating agentic reasoning, dynamic planning, and cross-modal feedback into one cohesive instructional brain, addressing the current gap in foresight-based educational AI systems [4].

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1. Introduction

In recent years, Intelligent Tutoring Systems (ITS) have become increasingly popular as tools for personalized digital learning [1]. These systems aim to mimic the guidance of a human tutor by providing instructional feedback, quizzes, and explanations. However, most existing ITS solutions remain reactive - they respond only after a student makes a mistake or completes an activity [5]. This limits their ability to guide learners through long-term educational goals or adjust teaching strategies based on evolving learner needs. The Agentic AI Core is designed to overcome this limitation by introducing proactive planning and decision-making into the tutoring process. Inspired by agentic AI principles, it simulates a goal-oriented tutor capable of breaking down learning objectives into multi-step instructional plans. Using advanced reasoning frameworks such as Tree of Thought and ReAct, the Agentic AI Core makes foresight-based decisions, dynamically adapts teaching strategies, and revises plans based on real-time feedback [2], [3]. This component acts as the central planning brain of a larger multimodal tutoring system, interacting with the Reinforcement Learning Engine, the LLM-driven content generator, and the Computer Vision module. It continuously tracks learner progress and emotional engagement using memory mechanisms, ensuring that the instruction remains coherent, personalized, and aligned with long-term goals. By integrating strategic planning, adaptive reasoning, and memory, the Agentic AI Core offers a step forward in developing emotionally aware, human-like tutoring systems [4], [6].

2. Background & Literature Survey

Background

Traditional Intelligent Tutoring Systems (ITS) rely heavily on rule-based logic or predefined decision trees to adapt to student performance [1]. While these systems can personalize content to some extent, they often lack the ability to plan instruction strategically, consider long-term goals, or simulate the pedagogical reasoning of a human tutor [5]. As a result, their interventions tend to be short-sighted, reactive, and insufficiently adaptive to changing learner needs.

Agentic AI offers a new paradigm for intelligent decision-making. Instead of passively responding to student inputs, agentic systems are designed to pursue complex, long-term

objectives using structured planning, memory, and self-directed reasoning [2]. These capabilities make Agentic AI particularly suitable for building tutors that not only respond to learners but also guide them intentionally across learning journeys [4].

Literature Survey

Recent frameworks such as ReAct, Tree of Thought, AutoGPT, and Voyager demonstrate how large language models (LLMs) can be embedded in agentic loops to simulate tool use, planning, and reflection [2], [3]. These systems combine reasoning chains, memory, and adaptive behavior to handle complex tasks like web automation, puzzle-solving, and goal-directed exploration. They highlight the emergence of AI agents that can operate autonomously in uncertain environments [7].

Despite their strengths, these systems show limitations when considered for tutoring. Most are tailored for narrow domains and lack integration with educational goals or pedagogy. They are often built around single-use tools without long-term planning depth. Memory integration is typically shallow, and emotional context or learner feedback is rarely considered [4]. Moreover, evaluations focus on task success, not on learning gains or instructional coherence [5].

In education, strategic reasoning must consider past learner progress, current affective state, and future instructional steps. Existing agentic frameworks fail to support this kind of cognitive-emotional adaptivity, and no standard agent architecture exists for ITS integration [1], [6].

2.1 Domain Overview

The Agentic AI Core operates within the domain of goal-oriented artificial intelligence, specifically focusing on agentic reasoning and planning in the context of intelligent tutoring systems (ITS). This domain blends elements from AI planning, reinforcement learning, large language models, and educational technology to simulate human-like instructional decision-making [1], [2].

Agentic AI refers to systems capable of autonomously pursuing complex, long-term objectives using memory, planning, and adaptive strategies [3]. Unlike reactive AI systems, agentic models continuously monitor progress, make decisions based on

environmental feedback, and revise their plans dynamically. In this context, the Agentic AI Core serves as the central "thinking brain" of the tutoring system, much like a human educator who evaluates a student's performance over time and adjusts their teaching approach accordingly [4].

Relevant subdomains include:

- Cognitive AI and planning frameworks (Tree of Thought, ReAct)
- Episodic and goal memory modeling
- Adaptive strategy selection
- Agent communication and coordination
- Multimodal learning system integration

The integration of agentic AI into education is a growing area of interest. By equipping ITS with planning and memory, the system moves from "responding" to "guiding." This domain enables the development of tutors that can create individualized learning trajectories, understand emotional and cognitive signals, and simulate human-like teaching strategies [4], [5].

2.2 Related Work

Several recent advances have laid the groundwork for agentic AI, demonstrating its potential for goal-driven reasoning and adaptive decision-making. Frameworks like ReAct, Tree of Thought, AutoGPT, and Voyager have shown how large language models (LLMs) can be embedded in agents to simulate planning, memory, and multi-step reasoning in dynamic environments [2], [3], [7].

- ReAct combines reasoning and acting by allowing agents to interleave thoughts with actions. This supports intermediate reasoning and more interpretable decision paths.
- Tree of Thought enables branching decision structures where the agent can evaluate multiple possibilities before acting, which improves planning depth.
- AutoGPT and Voyager are autonomous agents that use self-reflection and tool use to pursue open-ended goals across episodes.

While these systems demonstrate strong reasoning capabilities, they are primarily used in generic problem-solving (e.g., web navigation, coding, or simulations) and lack direct educational applications. They are not built to handle pedagogical goals, curriculum alignment, or the cognitive-emotional complexity of learners [4].

In educational AI, most ITS systems remain reactive. They deliver feedback based on quiz results or activity outcomes but do not simulate the strategic reasoning a human tutor would use [1], [5]. Some works propose using reinforcement learning to adapt strategies, but without a high-level planning layer, they lack foresight and coherence across learning sessions [6].

Furthermore, agentic systems evaluated in current literature are often tested on task benchmarks (HumanEval, WebShop), not on educational performance metrics like learning retention, instructional efficiency, or affective response [4], [5].

These gaps highlight the need for an Agentic AI Core that:

- Plans instructional strategies in alignment with long-term learning goals
- Adapts in real-time based on learner progress and feedback
- Integrates with other modules (CV, RL, LLM) for a unified tutoring experience

This component builds on the principles of agentic AI, but applies them to a new, unexplored domain: intelligent, emotionally-aware tutoring [4].

3. Research Gap

Identified Gaps and Motivation

Identified Gaps

Despite advances in ITS and agent-based frameworks, several key gaps remain unaddressed:

1. Lack of Strategic Planning in ITS

   Most current ITS operate reactively. They respond to mistakes or activity completions but do not simulate multi-step instructional planning. This results in fragmented and inconsistent learning guidance.

2. Shallow or Absent Memory Integration

   Existing systems struggle to maintain a coherent long-term record of learner progress. Without structured memory, they cannot adapt strategies based on past attempts, forgotten concepts, or prior emotional states.

3. No Real-Time Plan Revision

   Even when a high-level teaching plan exists, few systems revise it in real time based on changes in student engagement, confusion, or success. This leads to rigid instruction rather than responsive tutoring.

4. Disconnection Between Goal and Action

   Current agentic systems are task-driven, not goal-aligned. In an educational context, each action should serve a broader learning goal — but most AI systems lack that level of alignment and foresight.

5. Limited Application of Agentic AI in Education

   While agentic frameworks like ReAct or Tree of Thought are powerful, they have been mostly used in web automation, gaming, or tool-using agents - not in human tutoring contexts.

Motivation

This research is motivated by the belief that human-like tutoring behavior requires more than reactive feedback - it requires foresight, planning, and adaptation. A tutor should not only answer questions but also guide the learner strategically, adjusting based on memory and emotional cues.

The Agentic AI Core is designed to fill this gap by introducing goal decomposition, real-time plan revision, and cross-modal feedback integration. It aims to simulate a tutor who knows the destination, remembers the learner's path, and adjusts course thoughtfully - just like a skilled human educator.

By bridging agentic AI and educational intelligence, this component contributes toward building emotionally aware, foresight-driven, and pedagogically grounded AI tutors.

4.Research Problem

Most ITS today are reactive and operate without a structured sense of long-term instructional planning. They respond to individual student actions, such as wrong answers or low engagement, but fail to strategically guide learners through goal-aligned learning paths. This results in disjointed experiences that are unable to simulate the foresight, flexibility, and intentionality of a human tutor.

While recent advancements in agentic AI offer planning and memory capabilities, existing agentic agents are largely task-specific and have not been applied effectively in educational domains. They lack support for real-time emotional feedback, personalized goal tracking, and coherent pedagogical reasoning over time. Moreover, current implementations do not integrate multiple modalities such as visual engagement cues or reinforcement-based adaptation into a unified planning engine.

There is a clear need for a reasoning component that can:

- Decompose high-level learning goals into adaptable instructional plans
- Track student progress across sessions using memory
- Adjust strategies in real time based on cognitive and emotional feedback
- Collaborate with external modules (RL, CV, LLM) in a multimodal tutoring environment

This research focuses on designing and implementing an Agentic AI Core that addresses these needs by embedding strategic planning, memory-based adaptation, and foresight-driven tutoring into the heart of the system. The core challenge lies in building an

architecture that simulates human-like instructional reasoning while integrating with other AI modules to support personalized, emotionally-aware, and goal-aligned learning.

5. Objectives

5.1 Main Objective

To design and implement an Agentic AI Core that simulates a goal-oriented tutor capable of decomposing long-term learning objectives into adaptive instructional strategies using planning frameworks, memory-based tracking, and real-time feedback integration.

5.2 Specific Objectives

1. To develop a goal decomposition and planning module using frameworks like Tree of Thought or ReAct for multi-step instructional strategy generation.
2. To implement a memory-based tracking mechanism that retains learner progress, past actions, and feedback across sessions for coherent goal alignment.
3. To enable real-time plan revision based on inputs from the Computer Vision (CV) and Reinforcement Learning (RL) modules (e.g., emotional state, engagement, performance).
4. To integrate the Agentic AI Core with other system components (LLM, RL Engine, CV Module) through defined input/output protocols for coordinated instructional decisions.
5. To evaluate the system's effectiveness in terms of planning coherence, goal progression accuracy, and responsiveness to learner state changes.

6. Methodology

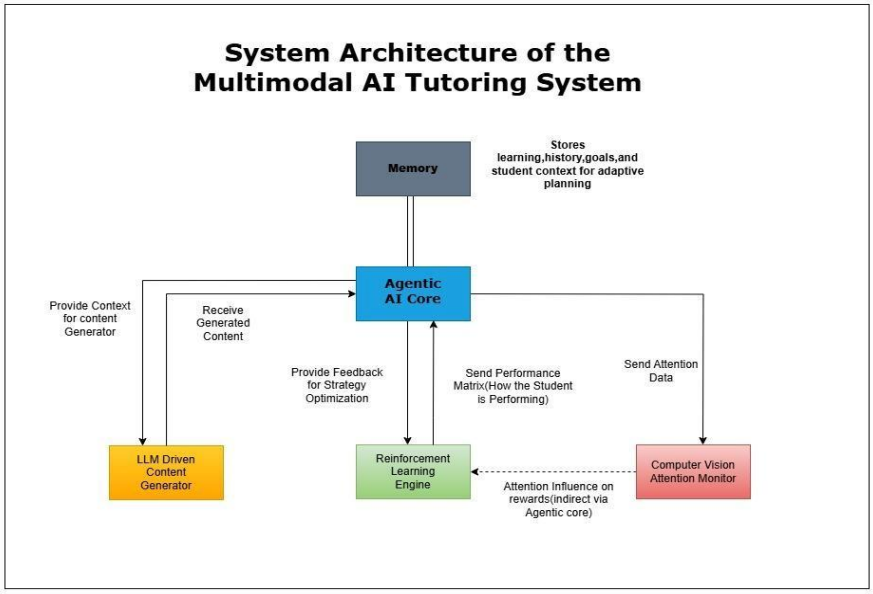
6.1 System Overview

The Agentic AI Core serves as the central planning and decision-making module of the multimodal AI tutoring system. It simulates the behavior of a human tutor by decomposing long-term learning goals into adaptive instructional strategies and dynamically revising them based on feedback.

At runtime, it interacts with:

- The Computer Vision module to receive real-time emotional and engagement cues
- The Reinforcement Learning Engine to receive or delegate optimized teaching actions
- The LLM module to generate explanations, quizzes, and feedback based on its instructional intent

The core uses frameworks like Tree of Thought and ReAct for planning, alongside a memory-based system that tracks the learner's progress and emotional trajectory. This creates a feedback-driven loop of planning, action, and observation, enabling personalized, goal-aligned instruction.



6.2 Tasks and Sub-Tasks

Task	Sub-Tasks
Planning Framework Implementation	• Evaluate and compare Tree of Thought vs ReAct • Implement planning strategy (select one or hybrid) • Test plan coherence for sample learning goals
Memory-based Goal Tracker	• Design memory schema for goal progress • Implement learner model with persistent state • Integrate memory updates after each instructional action
Real-Time Feedback Integration	• Define input format for emotional and engagement data from CV module • Set thresholds or triggers for plan revision • Simulate plan revision logic
Inter-Module Communication	• Define I/O formats for RL and LLM communication • Handle action delegation and feedback responses • Maintain coordination without conflict
Evaluation Module	• Design evaluation metrics: goal coverage, plan adaptability, learner state sensitivity • Run controlled simulations to assess core performance

6.3 Data Requirements

To build and test the Agentic AI Core, the following data will be required:

- Synthetic learner goal structures (e.g., course module breakdowns)
- Sample learner performance logs (simulated data for progress tracking)
- Emotional and attention signals from the CV module (mocked in early testing)
- Instructional actions & responses from RL and LLM modules (simulated or mocked for testing)
- Manually designed test cases for validating planning coherence and adaptivity

6.4 Responsibilities

As the owner of the Agentic AI Core component, my responsibilities include:

- Designing the planning architecture and selecting suitable frameworks (e.g., Tree of Thought, ReAct)
- Implementing memory-based learner tracking
- Handling real-time interaction logic between the agent and other modules (CV, RL, LLM)
- Defining input/output protocols for inter-module communication
- Evaluating the performance of instructional strategies generated by the core
- Ensuring the component aligns with the overall system goals and works cohesively with other components

7. Project Requirements

7.1 Functional Requirements

ID	Functional Requirement
FR1	The system shall decompose learning goals into multi-step instructional plans using Tree of Thought or ReAct.
FR2	The system shall maintain a memory model to track learner progress, feedback, and goal status.
FR3	The system shall revise instructional strategies in real time based on inputs from the CV and RL modules.
FR4	The system shall interface with the LLM module to trigger appropriate content generation actions.
FR5	The system shall log all planning decisions, learner states, and instructional actions for evaluation.

7.2 Non-Functional Requirements

ID	Non-Functional Requirement
NFR1	The planning engine shall respond to feedback and update plans within 1 second to maintain system fluidity.
NFR2	The memory model must be persistently stored and recoverable across sessions.
NFR3	The system shall be modular, allowing integration or replacement of planning strategies (ToT or ReAct).
NFR4	Data exchanged with other components must follow a defined JSON-based schema for consistency.
NFR5	The component must be testable in isolation using mock inputs from RL, CV, and LLM modules.

7.3 User Requirements

ID	User Requirement
UR1	The system shall allow researchers to define learning goals and sub-goals as input.
UR2	The user (developer or researcher) shall be able to view the generated instructional plan in a structured format.
UR3	The system shall provide logs or visual summaries of how plans are updated based on learner feedback.

7.4 System Requirements

Type	Requirement
Software	Python 3.x, JSON schema validator, basic simulation environment
Hardware	Any machine with standard CPU (agentic logic does not require GPU)
Libraries	Planning/decision-tree libraries, logging modules, mock input data generators
Interfaces	APIs for CV module (emotions, attention), RL module (feedback & action scores), LLM module (content triggering)

7.5 Tentative Use Cases

Use Case	Description
UC1	Tutor receives a new learning goal and generates a multi-step instructional plan.
UC2	Learner engagement drops; Agentic AI Core updates the plan with a simplified step.
UC3	RL recommends "give_hint" → Agentic AI Core triggers LLM to generate a hint.
UC4	Learner completes a task → Core updates goal status in memory and evaluates next step.
UC5	Agentic AI receives emotion input (e.g., frustration) and decides to revise plan tone or difficulty.

7.6 Tentative Test Cases

Test Case	Input	Expected Output
TC1	Learning goal: “Understand Algebra”	Plan with 3–5 instructional steps
TC2	Emotion input: "frustrated", engagement = 0.3	Revised plan with easier step or visual explanation
TC3	RL recommends "repeat_concept"	Agentic AI schedules concept review and triggers LLM
TC4	Learner completes quiz with low score	Goal progress is updated; plan adjusts future difficulty
TC5	Simulated disconnection from CV module	Plan continues based on memory and default assumptions

8. References

[1] V. J. S. a. J. Psotka, "Wikipedia," December 2024. [Online]. Available: https://en.wikipedia.org/wiki/Intelligent_tutoring_system. [Accessed August 2025].

[2] S. G. Y. e. al., "ReAct: Synergizing reasoning and acting in language models," 2023. [Online]. Available: https://arxiv.org/abs/2305.10601. [Accessed August 2025].

[3] S. Z. e. al., "Tree of Thoughts: Deliberate problem solving with large language models," 2023. [Online]. Available: https://arxiv.org/abs/2305.10601. [Accessed August 2025].

[4] K. R. K. a. A. T. Corbett, "Cognitive tutors: Technology bringing learning sciences to the classroom," 2006. [Online]. Available: https://ieeexplore.ieee.org/document/5376142. [Accessed August 2025].

[5] B. P. Woolf, "Building intelligent interactive tutors: Student-centered strategies for revolutionizing e-learning," 2009. [Online]. Available: https://www.researchgate.net/publication/220814816_Intelligent_Tutoring_Systems_New_Challenges_and_Directions. [Accessed August 2025].

[6] J. R. A. e. al., "Cognitive tutors: Lessons learned," 1995. [Online]. Available: https://ieeexplore.ieee.org/document/4913231. [Accessed August 2025].

[7] J. W. e. al., "Voyager: An open-ended embodied agent with large language models," 2023. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10849561. [Accessed August 2025].

[8] A. M. e. al., "Evaluating the effectiveness of intelligent tutoring systems in real-world settings," 2017. [Online]. Available: https://ieeexplore.ieee.org/document/8095141. [Accessed August 2025].